Year 13 Further Maths - Poisson distribution

Similar questions - Modelling and goodness-of-fit tests

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 The average numbers of cars, vans and buses passing a point on a road in 20 minutes are 300, 60 and 12 respectively.

(a) Assuming that the numbers of each type of vehicle are independent Poisson variables, find the probability that the total number passing in a random 20-minute period is at least 390. **[3]**

(b) Buses run to a fixed timetable. Explain why the Poisson distribution is not a good model for the number of buses passing in a fixed time. **[1]**

2 The numbers of CD players sold in a shop in three consecutive weekends were 3, 9 and 6. It may be assumed that sales occur randomly and independently. The number sold in a randomly chosen weekend is X.

(a) Comment on how appropriate the Poisson distribution is as a model for X. **[2]**

Now assume $X \sim \text{Po}(6)$.

(b) Find (i) $\text{P}(X = 5)$, (ii) $\text{P}(X \geqslant 9)$. **[4]**

(c) The number of sound systems sold in a weekend has distribution $\text{Po}(4.5)$. Find the probability that the total sold in a weekend is between 9 and 13 inclusive. **[3]**

(d) State an assumption needed for part (c), and give a reason why it may not be valid. **[2]**

3 A breakdown service models the number of punctures R in a 24-hour period by $\text{Po}(3.1)$.

(a) State a further assumption needed. **[1]**

(b) State the standard deviation of R. **[1]**

(c) Find the probability that in 7 days there are at least 25 punctures. **[3]**

(d) The number of flat batteries in 24 hours is modelled by $\text{Po}(1.2)$, independently. A "bad" day is one on which the total number of punctures and flat batteries exceeds 5. Find the probability that a day is bad. **[3]**

(e) In 100 days, 12 days were found to be bad. Comment on whether this casts doubt on the models. **[3]**

4 A researcher believes that the number of calls received in a 10-minute interval at a call centre is Poisson. The number of calls in 60 random intervals is summarised below.

Calls	0	1	2	3	4	$\geqslant 5$
Frequency	10	18	17	9	4	2

For the table, take the mean of the sample as 1.75 (taking each interval with 5 or more calls as exactly 5) and the variance as 1.65.

(a) Comment on what the mean and variance suggest about the model. **[1]**

(b) Explain why it is necessary to combine cells in the test. **[1]**

(c) Show that the expected frequency for 2 calls is 15.97 (2 d.p.). **[2]**

(d) Carry out a goodness of fit test at the 5% level. **[6]**

5 Bird-watchers study the number N of separate bursts of song by a species in 5-minute periods. They assume that a burst of song can be treated as a single event.

(a) State two further assumptions needed for N to be modelled by a Poisson distribution. **[2]**

(b) The mean and variance of the results from 60 periods are 3.08 and 3.57. Explain what this suggests about a Poisson model. **[2]**

The frequencies are shown below.

n	0	1	2	3	4	5	6	7	8
Frequency	4	9	12	12	10	6	4	2	1

(c) State hypotheses for a goodness of fit test. **[1]**

(d) Using mean 3.083, determine the contribution to the test statistic when $n = 3$. **[3]**

(e) Combining the cells for $n \leqslant 1$ and $n \geqslant 6$ the test statistic is 0.93 (2 d.p.). Complete the test at the 5% level. **[3]**

(f) It is known that these birds are more likely to sing when other birds of the same species are singing. Explain how this affects the validity of the Poisson model. **[1]**

Question 1
(a) Total $\sim$ Po(372); $\mathrm{P}(T \geqslant 390) = 1 - \mathrm{P}(T \leqslant 389) = 0.182$
(b) Events must occur randomly; buses arrive at regular intervals so are not random

Question 2
(a) Mean 6 and sample variance 9: the variance is larger than the mean, which suggests that Poisson may not be appropriate, but with only three observations it is hard to say
(b)(i) 0.161 (ii) $1 - \mathrm{P}(X \leqslant 8) = 0.153$
(c) Total $\sim$ Po(10.5): $\mathrm{P}(9 \leqslant T \leqslant 13) = \mathrm{P}(T \leqslant 13) - \mathrm{P}(T \leqslant 8) = 0.546$
(d) The two sales counts must be independent; a customer might buy both, or a promotion might raise both

Question 3
(a) Punctures occur independently and at a constant average rate
(b) $\sqrt{3.1} = 1.76$
(c) Seven days $\sim$ Po(21.7); $\mathrm{P}(\geqslant 25) = 0.266$
(d) Total $\sim$ Po(4.3); $\mathrm{P}(> 5) = 0.263$
(e) Expected number of bad days = 26.3; $\mathrm{P}(\leqslant 12)$ for $\mathrm{B}(100, 0.263)$ is about 0.0004: very unlikely, so there is strong evidence that the models are not valid (e.g. rates vary by season or are not independent)

Question 4
(a) Mean $\approx$ variance, which is consistent with Poisson
(b) Some expected frequencies are less than 5 (that for 4 is 4.07 and for 5 or more is 1.97)
(c) $60\,\mathrm{P}(X = 2) = 60 \times \dfrac{\mathrm{e}^{-1.75} 1.75^2}{2!} = 15.97$
(d) H_0: Poisson is a suitable model; H_1: it is not. Combining $r \geqslant 4$: $E = 10.43,\ 18.25,\ 15.97,\ 9.31,\ 6.05$; contributions $0.017,\ 0.003,\ 0.067,\ 0.011,\ 0.0004$, $\chi^2 = 0.099$. Degrees of freedom $5 - 1 - 1 = 3$, critical value 7.815. $0.099 < 7.815$ so do not reject H_0

Question 5
(a) Bursts occur independently of one another; bursts occur at a constant average rate; two bursts cannot occur at the same time
(b) The variance is greater than the mean, which suggests more variability than a Poisson model allows (overdispersion)
(c) H_0: a Poisson distribution is a suitable model; H_1: it is not
(d) $E = 60\,\mathrm{P}(N = 3) = 60 \times \dfrac{\mathrm{e}^{-3.083} 3.083^3}{3!} = 13.43$; contribution $\dfrac{(12 - 13.43)^2}{13.43} = 0.152$
(e) Cells: $n \leqslant 1, 2, 3, 4, 5, \geqslant 6$ so $6 - 1 - 1 = 4$ degrees of freedom; critical value 9.488. $0.93 < 9.488$ so there is no evidence against a Poisson model
(f) Bursts are then not independent, so the Poisson assumption fails, and this may explain the larger variance